

Chapter 52 — Ectopic Ureter, Ureterocele, and Ureteral Anomalies

CLASSIFICATION AND ANATOMIC DESCRIPTION

General Patterns

- Ureteral developmental abnormalities — ectopic ureters and ureteroceles — are a **large component of clinical urology and continue to challenge management despite wide recognition and well-defined surgical strategies**.
- The difficulty arises from a **wide spectrum of clinical significance and a variable pattern of presentation**, which is why a working grasp of normal and abnormal lower urinary tract embryology is required rather than a list of operations.
- Ectopic ureter and ureterocele are taught as distinct entities but **share many features and probably share underlying developmental mechanisms**, with slight variations producing the obvious differences in appearance.
- Management is approached similarly with only slight variation; **clinical cases exist that lie between the two entities, suggesting a developmental continuum** rather than two separate diseases.

Concept — *The two entities are best treated as one developmental defect of ureterotrigonal connection expressed along a continuum; the differences in appearance dictate the technical variation in management, not the strategy.*

Ectopic Ureter

- Definition: **any ureter, single or duplex, that does not enter the trigonal area of the bladder**.
- A ureter merely near the bladder neck is **not technically ectopic** — **ectopia requires entry at or distal to the bladder neck**.
- In a duplex system the ectopic ureter is **inevitably the upper pole ureter**, because it buds **later and more cephalad from the mesonephric duct** than the lower pole, with later incorporation into the developing urogenital sinus.
- Association with a **Gartner duct cyst** — the remnant of the wolffian duct from which the ureter buds, which may itself show cystic dilation. The duct **runs parallel alongside the vagina** (the müllerian structure); rupture of the cystic duct into the vagina establishes communication. **This is the anatomical basis of incontinence in the female**.
- Male presentation therefore **never includes incontinence**; instead hydronephrosis, flank pain, urinary tract infection (UTI), or epididymitis/orchitis.
- **Zinner syndrome** = ectopic ureter with **ipsilateral renal absence and seminal vesicle cyst**. A retrograde injection study may demonstrate the ureterocele or ureter communicating with the seminal vesicle and vas deferens, the two structures joining just above the seminal vesicles.
- Single-system ectopic ureters and ureteroceles may mimic the ectopic upper pole but may also present with **an apparently absent kidney** — usually a failure of ultrasound to identify a small, poorly functioning unit, prompting **computed tomography (CT) with contrast or renal function studies**.
- Either single or duplex systems may show **severe hydroureteronephrosis reflecting distal obstruction**; obstruction may be severe enough to have **impaired renal development to the point of non-function**.
- **Bilateral single-system ectopic ureters** (rare) are associated with a **hypoplastic bladder, poorly developed trigone and bladder neck, and bilateral renal dysplasia**.

COMPARISON — ECTOPIC URETER: THE GOVERNING SEX ASYMMETRY

	Female	Male
Insertion sites	Bladder neck to perineum; vagina, uterus, rectum	Always above the external sphincter or pelvic floor ; vas deferens, seminal vesicle, ejaculatory duct
Relation to müllerian structures	Indirect, via the Gartner duct running alongside the vagina	Not applicable; insertion is into wolffian derivatives
Continence	Continuous incontinence is the classic presentation	Incontinence never occurs
Typical presentation	Continuous wetting, perineal discharge, UTI	Epididymitis/orchitis, UTI, hydronephrosis, flank pain
Eponymous association	Gartner duct cyst	Zinner syndrome (renal agenesis + seminal vesicle cyst)

- The table exists to fix **one variable — the level of insertion relative to the external sphincter** — as the generator of every clinical difference between the sexes.
- The trap: treating incontinence as a feature of "ectopic ureter" generally, when it is a feature of **female** ectopic ureter only.

Concept — *Everything downstream follows from which duct system the ectopic ureter joins relative to the external sphincter: the female ureter can open distal to the sphincter and therefore leaks continuously; the male ureter, confined to wolffian structures above the sphincter, cannot leak and must present as infection or obstruction instead.*

KEY POINTS: ECTOPIC URETER

- An ectopic ureter is **any ureter, single or duplex, that does not enter the trigonal area of the bladder**.
- In a duplex system it is **inevitably the upper pole ureter**, because budding from the mesonephric duct occurs later and more cephalad than the lower pole bud.
- In females it may enter **anywhere from the bladder neck to the perineum, and into the vagina, uterus and even rectum; continuous wetting is a classic symptom**.
- In males it **always enters above the external sphincter or pelvic floor**, usually into wolffian structures; **the presentation does not include incontinence**.

Ureterocele

- Definition: **a version of the ectopic ureter with cystic dilation of the distal ureter, sited either wholly within the bladder or spanning the bladder neck and urethra**.
- As with ectopic ureter, may be **single or duplex; in duplex systems always the upper pole**.
- Ureteroceles **may extend into the urethra but never form entirely within it, and never attach to wolffian ductal structures** — the point of separation from the ectopic ureter.
- The **most clinically useful classification separates intravesical from extravesical (ectopic) ureteroceles**.

Intravesical Ureterocele

- **Entirely within the bladder and above the bladder neck**.
- Includes the so-called "simple" ureterocele of the adult with mild or no upper tract dilation; **the term "simple" should be discouraged in favour of "single intravesical ureterocele"**.

- **Intravesical ureteroceles can still obstruct**, producing severe hydronephrosis and clinically significant sequelae.

Ectopic (Extravesical) Ureterocele

- Some portion is **permanently situated at the bladder neck or urethra**. The orifice may be in the bladder, at the bladder neck, or in the urethra.
- **This must be distinguished from an intravesical ureterocele that merely prolapses into the urethra with voiding.**

Morphological Variants and Functional Grading

- **Stephens classification:** cecoureterocele, stenotic, sphincteric, sphincterostenotic, blind, and non-obstructed.
- **Cecoureterocele — the critical subgroup:** the **orifice lies within the bladder, but the ureterocele cavity extends submucosally beyond the bladder neck into the urethra**, as a long tongue beneath the ureteral submucosa; the orifice communicating with the bladder lumen is **large and incompetent**. Often **not characterised preoperatively**, and its complexity creates surgical difficulty, particularly with endoscopic incision.
- **Ureterocele disproportion** (non-obstructive ureterocele with duplication): duplex kidney in which the **upper pole is non-dilated and so dysplastic that it is not detected on most imaging**, with **no ureteric dilation beyond the bladder** — a typical-appearing ureterocele in the bladder with an apparently normal ipsilateral kidney.
- **Churchill functional classification** grades by **impact on the upper urinary tract across all renal units**: upper pole alone in jeopardy → entire ipsilateral kidney threatened → contralateral system also at risk through reflux or **bladder outlet obstruction (BOO)**. The last group comprised roughly a quarter of the reported cases.
- Ureterocele type shows **some association with therapeutic outcome, but the predictions have not proved as reliable as first reported**.

COMPARISON — THE THREE URETEROCELE CLASSIFICATION SYSTEMS

	Glassberg	Stephens	Churchill
Basis	Anatomical position relative to the bladder neck	Morphology of orifice and cavity	Functional impact on renal units
Categories	Intravesical vs ectopic (extravesical)	Cecoureterocele, stenotic, sphincteric, sphincterostenotic, blind, non-obstructed	Upper pole only / whole ipsilateral kidney / contralateral also at risk
Clinical value	Most useful in practice; predicts outcome of endoscopic incision	Identifies the cecoureterocele , the surgically dangerous variant	Frames the extent of the threat, including BOO
Weakness	Binary; ignores cavity morphology	Largely descriptive	Outcome prediction unreliable

- The table is for choosing **which classification answers which question**: position predicts endoscopic success, morphology predicts surgical difficulty, function predicts what is at stake.
- The pattern across the rows: **all three describe the same lesion at increasing distance from the orifice** — the orifice itself, then the cavity, then the kidneys.
- The trap: using Stephens categories clinically when **only the cecoureterocele subgroup carries management weight**.

COMPARISON — ECTOPIC URETER VERSUS URETEROCELE

	Ectopic ureter	Ureterocele
Distal morphology	Non-dilated or dilated ureter opening ectopically	Cystic dilation of the distal ureter
Position	At or distal to the bladder neck; may reach perineum, vagina or rectum (female) or wolffian ducts (male)	Within the bladder or spanning the bladder neck and urethra; never entirely within the urethra
Wolffian attachment	Yes — characteristically in the male	Never
Incontinence	Yes, in females	Not when untreated
Duplex association	Upper pole	Upper pole

- The two discriminating rows are **wolffian attachment and untreated incontinence**; the rest of the table is shared ground.
- The trap: assuming the entities are separated by the presence of dilation. They are separated by **where the distal segment sits and what it is attached to**.

Concept — *The ureterocele is the same distal ureteric maldevelopment with a retained, distensible distal segment; the defining questions in both conditions are always the same — where is the orifice relative to the bladder neck, and what does the distal segment do to the bladder outlet.*

KEY POINTS: URETEROCELE

- A ureterocele is a version of the ectopic ureter with **cystic dilation of the distal ureter, located within the bladder or spanning the bladder neck and urethra**.
- Several classification systems exist, but **the most useful distinction for clinical practice separates intravesical from extravesical ureteroceles**.
- In a **cecoureterocele** the orifice of the affected ureter is **within the bladder**, but the cavity **extends submucosally beyond the bladder neck into the urethra**.

Embryology and Aetiology

- The **specific mechanisms remain undefined**, but understanding of normal and abnormal ureterotrigonal development is the likely source of insight.
- Pathogenesis: **ureteral ectopia, with or without ureterocele, produces renal maldevelopment through defective ureterotrigonal connections** — the renal lesion is a consequence of the connection defect, not a separate event.
- A growing number of **genetic pathways critical to distal ureter connection** have been identified in humans and rodents; mouse models have clarified how proper distal connections are generated.
- **Classic model**: the **common nephric duct (CND) forms the trigone**.
- **Revised model**: the **CND regresses by apoptosis**, and the **trigone forms from endoderm, like the bladder**.
- **Chwalle's classic explanation of ureterocele formation**: distal ureteral expansion results from **failure of rupture of a distal membrane at the ureteral orifice — the Chwalle membrane**.
- Because the **ureteral bud derives from the wolffian duct**, the ectopic ureter **does not insert directly into müllerian structures** (vagina, cervix, uterus) but is **associated with them through the Gartner duct**, the wolffian remnant running alongside the mature müllerian structures.

- **Weigert-Meyer rule:** the **inverse relationship of duplex ureteral orifices** — the ectopic or ureterocele-bearing **upper pole orifice is caudal to the lower pole orifice**.

Concept — *Two mechanisms generate the whole chapter: a mis-timed, mis-sited ureteral bud, which fixes the orifice position and therefore the Weigert-Meyer inversion and every ectopic site, and a failure of distal orifice canalisation, which fixes the ureterocele. Renal dysplasia is downstream of both, not parallel to them.*

CLINICAL PRESENTATION

Imaging

Prenatal Detection

- The majority of ureteroceles and ectopic ureters are detected by prenatal ultrasound, often only through the identification of hydronephrosis; the **specific diagnosis is frequently not made until after birth**.
- Prenatal patterns are **identical to postnatal ultrasound patterns but more easily misinterpreted**.
- A duplex system is difficult to identify unless one moiety is dilated.
- An upper pole "cyst" in a fetus should be read as upper pole hydronephrosis until proven otherwise.
- Minor hydronephrosis may be missed, though **change over time prompts recognition**; expected ureteric dilation may not be obvious but **can often be traced down to the bladder**.
- The bladder must be inspected in every such case to identify a ureterocele, and it may be necessary to **await bladder filling** — a large ureterocele can fill the bladder and mask its own presence.
- Record upper pole parenchymal thickness and echogenicity — these findings inform later clinical decisions.
- The ectopic ureter looks **identical to a ureterocele except at bladder level**: dilated upper pole, tortuous ureter, **no intravesical component**. A large ectopic ureter may impinge on the bladder and appear intravesical — a **pseudoureterocele**. Recognition requires an experienced maternal-fetal sonographer.
- With a tentative diagnosis, **evaluate the other renal units and the bladder**: ipsilateral lower pole or contralateral dilation suggests **reflux, or less commonly obstruction**.
- **BOO from a ureterocele can obstruct all renal units; oligohydramnios from this cause is very unusual but does occur**. Contralateral renal dysplasia may coexist with reduced amniotic fluid.
- The need for prenatal intervention or early delivery is exceptional and unlikely to confer significant benefit.
- A **single-system** lesion is suggested by **dilation of the entire kidney and ureter**; this may be indistinguishable from obstructive megaureter or severe reflux, but **the distinction has little immediate prenatal consequence** and is made postnatally.

Concept — *Prenatally the diagnosis is made at the bladder, not at the kidney — upper tract appearances of ectopic ureter and ureterocele are interchangeable, so the entire diagnostic yield lies in imaging a filled bladder.*

Incidental Postnatal Detection

- Sometimes diagnosed from **hydronephrosis found on imaging performed for unrelated reasons** — trauma, abdominal pain.
- Some studies performed for general abdominal pain **not thought to be renal prove to be of renal origin**.
- A **dilated ureter may not be recognised as such**: presumed ovarian cysts have proved to be markedly dilated ureters. **Always include the dilated ureter in the differential of a cystic pelvic structure**.

Infection

- Infection remains a **significant reason for clinical presentation of both entities, at any age, with a highly variable pattern.**
- **Generalised urosepsis may be the presenting scenario, and renal-bladder ultrasonography will usually provide the diagnosis.**
- It has been reported that **acute ultrasound is unnecessary in a child with febrile UTI and a normal prenatal ultrasound after 30 to 32 weeks**; the authors consider this **tenuous given the variable quality of prenatal imaging**, and hold that **ultrasound should be obtained in all children with urosepsis.**
- In febrile UTI **there is rarely urgency if the child is responding clinically**; the value of early detection is the **possibility of a simple drainage procedure.**
- **Sepsis not responding rapidly to appropriate therapy → urgent transurethral incision (TUI) of an infected ureterocele, or percutaneous or open drainage of an ectopic ureter**; open drainage in this setting is usually an **end ureterostomy.**
- Ectopic ureters frequently present **less acutely, with ongoing low-grade fever and periodic spikes.**
- **A negative urine culture does not exclude infection** — the infected ectopic system may not drain into the bladder.
- In young females, parents may describe **purulent discharge from the perineum**; this pattern should prompt ultrasound, usually revealing a dilated upper pole or whole system.
- **The unobstructed ectopic ureter rarely presents with infection but can present with incontinence.**
- In young males the subacute pattern occurs, but **epididymitis is the more common presentation** — true bacterial epididymitis or infected urine.
- **In any young male with suspected bacterial epididymitis, perform upper tract ultrasonography to exclude an abnormality.**
- Older males present the same way, often **after recurrent episodes of epididymitis before the cause is found**; the reason for the delay is unclear.

Concept — *Infection in these anomalies is infection in a compartment that does not communicate with the bladder; that single fact explains the negative culture, the perineal discharge, the epididymitis, and the need to image rather than treat empirically.*

Incontinence

- Incontinence may be caused by an ectopic ureter in a female but never in a male; untreated ureteroceles are not associated with incontinence.
- The toilet-trained female with verified continuous urinary leakage must be evaluated for an ectopic ureter.
- Imaging may not immediately detect it because the affected moiety may not be dilated — suspicion must be driven by history and occasionally physical examination.
- Before toilet training the pattern is hard to detect, though **some parents note persistent dribbling during changing**; in early training the wetting is often **falsely attributed to lack of motivation.**
- **The characteristic history is persistent low-volume dampness at all times, day and night, that does not relent, sometimes worsening with activity.**
- **The discriminating question: can the child be dry for 30 to 60 minutes? With an ectopic ureter the answer is no.**
- Caution in questioning: **parents of children wetting for other reasons will also say the child is always wet when dry intervals in fact exist.**

- **Rarely the leakage is intermittent**, perhaps from intermittent leakage through a **membrane of the Gartner duct**.
- In older undiagnosed children, symptoms may be misattributed to **dysfunctional voiding, laziness, or even sexual abuse**; **seek voiding postponement, posturing and constipation** to assess whether voiding dysfunction explains the wetting.

Pain

- **Pain is uncommon with either entity.**
- Exceptions: **pain from acute infection, episodic obstruction of the ectopic ureter, and bladder pain from an obstructing ureterocele.**
- **Intermittent drainage of an ectopic ureter** in older children: **abdominal pain followed by perineal drainage of urine or purulent material**, presumed to reflect accumulation behind an obstructed orifice with subsequent release.

Prolapse

- **Ureterocele prolapse is unusual but distinctive** and still confuses clinicians.
- Appearance: **smooth, congested, mucosa-covered interlabial mass protruding from the urethra, distinct from the vagina**, often with difficulty voiding.
- **It is not lobulated — that appearance would suggest sarcoma botryoides.**
- **Bladder ultrasound usually confirms the diagnosis**, with renal images supporting it. The accompanying voiding cystourethrogram characteristically shows the ureterocele prolapsing into the urethra with **lower pole reflux and the classic "drooping lily" appearance** of the lower pole displaced by the dilated upper pole.

Late Presentation

- Presentation in the **teenager or adult** is reported, usually with **infection or abdominal pain, rarely incontinence**.
- **The non-obstructing ureterocele, often single-system, is well recognised in the adult — frequently containing a stone.**
- **Vaginal wall prolapse** has been associated with an ectopic ureter; **ureterocele prolapse after incision** has been reported in adults.
- **There is usually little or no function in the affected moiety.** Upper pole heminephrectomy in duplex systems has been the mainstay, but **upper-to-lower pole ureteroureterostomy may also be considered.**

Concept — *Presentation is determined by sex, by whether the system drains into the bladder, and by whether the distal segment obstructs the outlet — continuous wetting, occult infection, and an interlabial mass are three expressions of the same anatomical question.*

KEY POINTS: CLINICAL PRESENTATION

- The majority are **detected by prenatal ultrasound on the basis of hydronephrosis**, although a specific diagnosis is often not made until after birth.
- Ectopic ureters frequently present with a **less acute pattern of low-grade fever with periodic spikes**; **urine cultures may be negative because the infected system does not drain into the bladder.**
- Young males with an ectopic ureter may present subacutely but **often have associated epididymitis.**
- **Incontinence may be caused by an ectopic ureter in young females but not in young males, and the toilet-trained female with verified continuous leakage must be evaluated for one.**

- **Ureterocele prolapse** is an unusual but distinctive sign — a **smooth, congested, mucosa-covered interlabial mass**, often with difficulty voiding.

EVALUATION

Management rests on a thorough assessment of the affected anatomy and of the functional implications of the condition, together with the operative and postoperative care anticipated. Both renal and bladder function are affected by the developmental abnormality and may ultimately require complex surgical intervention; **voiding, bladder emptying, vesicoureteral reflux (VUR), renal drainage and kidney function** are all elements of the management plan.

Anatomic Assessment

Physical Examination

- **Provides invaluable information in both conditions.**
- **The prolapsed ureterocele is diagnostic and dramatic but unusual**; further anatomical study is still necessary.
- **Detecting an ectopic perineal orifice in a child is unusual but highly informative** as to the character of the problem and the best treatment — **a search for it is worthwhile in any child with continuous wetting or a known leaking ectopic ureter**. When found it lies **between the urethral orifice and the vagina** and can be cannulated with an angiocatheter.
- **Rarely a dilated Gartner duct cyst is detected on careful perineal examination**; orifices or cystic structures found can then be **imaged by contrast injection**.
- **The dilated upper pole may be palpable in a relaxed infant**, but rarely in an older child.

Ultrasonography

- **Usually provides the anatomic diagnosis and permits inference of functional status.**
- Typical findings: **dilated upper pole with ureteric dilation, or a dilated single system.**
- Parenchymal character is **more readily seen postnatally**; if it appears healthy, **the true need for functional upper tract imaging is controversial** — it may be argued unnecessary if it will not change treatment.
- **The difference between functioning and non- or poorly functioning upper poles is usually apparent.**
- **Above the level of the bladder there are no features that distinguish a dilated ectopic ureter from a ureterocele** — so the **bladder images are critically important**, because management differs greatly.
- **Ureterocele on bladder views: a thin-walled cystic dilation within the bladder that does not extend beyond the bladder walls.**
- Laterality is usually apparent but **may appear midline if the ureterocele is large**; a **lobulated ureterocele may appear as two structures**, but careful examination should demonstrate communication.
- **Extension of a cecoureterocele into the bladder neck is difficult to visualise** — non-visualisation is not diagnostic.
- **Confusing patterns:** a large ureterocele may **completely fill the bladder with no urine visible**; a **full bladder with an effaced ureterocele may be mistaken for an ectopic ureter** — observation over time avoids both errors.
- **Pseudoureterocele:** a very dilated ectopic ureter impinging on the bladder wall, distinguished because the wall is **bilaminar and much thicker than a ureterocele wall**, and the **lumen extends well outside the bladder walls**.

Voiding Cystourethrogram

- The voiding cystourethrogram (VCUG) gives excellent evaluation of the bladder, urethra and often the distal ureters.
- Reflux is most often into the lower pole of the duplicated kidney, identified by the orientation of the calyces; reflux may also be contralateral and/or into the ureterocele itself.
- Because the presence of reflux determines initial treatment for some practitioners, VCUG is frequently obtained.
- Where a ureterocele is significantly obstructing on ultrasound and early incision is planned, the initial VCUG may be optional; with BOO or severe bilateral upper tract obstruction, VCUG findings would not alter treatment and transurethral puncture can proceed on ultrasound alone.
- After incision, a post-puncture VCUG should be considered, as ipsilateral or contralateral reflux guides subsequent decisions.

Magnetic Resonance Imaging

- Magnetic resonance (MR) urography gives the most detailed images of the affected tract.
- Magnetic resonance imaging (MRI) in young children usually requires sedation or anaesthesia and typically adds nothing to cheaper methods.
- Its value rests with patients in whom other imaging cannot define complex anatomy: a duplex system with a non-dilated upper pole, one with massive dilation obscuring whether duplication exists, or one with significantly distorted anatomical relationships.
- Where such delineation is needed, MRI adds functional information as well.

COMPARISON — IMAGING MODALITIES IN ECTOPIC URETER AND URETEROCELE

Modality	What it answers	When it is the right test	Limitation
Ultrasound	Anatomic diagnosis; inferred function; bladder character	First study in every case, prenatal and postnatal	Cannot distinguish ectopic ureter from ureterocele above the bladder
VCUG	Reflux — lower pole, contralateral, into the ureterocele; bladder base and urethra	Before and after intervention; assessing eversion and prolapse	Optional where obstruction alone dictates incision
DMSA	Differential renal function — gold standard	Deciding preservation versus removal	No criteria define "normal" upper pole function
MAG3 diuretic renogram	Function plus drainage	Candidates for observational management	Limited value for the dilated ectopic ureter, rarely observed
MR urography	Most detailed anatomy plus function	Complex or undefined anatomy only	Sedation or anaesthesia; expensive; usually adds nothing
Intravenous pyelogram	Anatomy with qualitative function	Uncertain anatomy; non-dilated ectopic ureter with incontinence	Qualitative only; a poor baseline study

- The table is for matching a question to a study, not for ranking studies: each row answers something none of the others does.
- The pattern: anatomy is settled by ultrasound and, exceptionally, MRI; reflux by VCUG; function only by nuclear imaging.

- The trap: ordering the most detailed study rather than the one that changes management — **MRI and the intravenous pyelogram are both reserved, not routine.**

Functional Assessment

Renal Function — Nuclear Imaging

- **Radionuclide renal imaging remains the gold standard for renal functional assessment**, usually best afforded by **dimercaptosuccinic acid (DMSA) imaging**, though obtainable with a **mercaptoacetyltriglycine (MAG3) diuretic renogram**.
- **The affected upper pole is the principal focus, but the health of the other moieties must be determined**, particularly with lower pole reflux or hydronephrosis of any unit.
- **There are no criteria defining "normal" function of an upper pole associated with an ectopic ureter or ureterocele, and no objective parameters for the level of functional contribution worth preserving.**
- Functional assessment therefore offers only the **dichotomy of some function versus no function** — generally clear, but **always with ambiguous cases**.
- **Preoperative assessment cannot predict function after decompression.**
- Where **observation is contemplated**, a diuretic renal scan replaces the DMSA scan, giving **both function and drainage**; this has value where **drainage is adequate and hydronephrosis has been reported ultimately to resolve**.
- **A dilated ectopic ureter is only rarely suitable for observation**, so obstruction assessment has limited value there.

Renal Function — Intravenous Pyelogram

- The intravenous pyelogram (IVP) may occasionally give adequate functional information with anatomical delineation without the complexity of MRI.
- **Functional assessment is qualitative only**, and the IVP is a **less useful baseline study except where anatomy is uncertain or an ectopic ureter with no dilation is suspected in the setting of incontinence**.

Bladder Function

- **The bladder is not the principal focus but its function cannot be ignored.**
- Impairment may arise from **BOO by a ureterocele**, or impaired continence from **inadequate bladder neck function** with either entity.
- **In bilateral single-system ectopic ureters, bladder development may be severely impaired because of the absence of fetal bladder filling.**

Ultrasound imaging interpretation

- A simple bladder ultrasound is very useful: **waiting for filling and emptying yields bladder emptying efficiency and wall thickness**, the latter indicating possible outlet obstruction.
- **The character, location and size of the ureterocele can usually be assessed, and the ectopic nature of a dilated ureter is sometimes identifiable.**

Ureterocele and the bladder outlet

- **The appearance of the bladder base with filling and voiding on VCUG guides therapeutic decisions: significant eversion indicates a weak trigonal floor more likely to require surgical repair.**
- **Prolapse into the urethra, or demonstration of a cecoureterocele, is important in planning and performing TUI and may better predict the need for secondary surgery.**
- **A patulous bladder neck with either lesion may be complicated by later incontinence, though continence is hard to predict at presentation.**

- **Cecoureteroceles are a particular risk factor for incontinence;** bladder neck ectopic ureters with wide-open bladder necks have been observed to be **incontinent after resection of the distal ectopic ureter and external bladder neck reinforcement.**
- Such patients **may benefit from more aggressive bladder neck tightening at definitive resection, accepting the risk of disturbing what may be normal bladder neck function; videourodynamic evaluation may help.**

COMPARISON — THE FOUR VCUG DESCRIPTIVE TERMS FOR A URETEROCELE

Term	Definition
Effacement	Flattening of the ureterocele with bladder filling
Prolapse	Movement of the ureterocele into the urethra
Eversion	Diverticulum-like protrusion of the bladder base and trigone with filling and voiding
Intussusception	Protrusion of the ureterocele back into its own ureter

- The four terms separate **what the ureterocele does to itself (effacement, intussusception) from what it does to the outlet (prolapse) and what the bladder base does around it (eversion).**
- Only two carry management weight: **eversion signals a weak trigonal floor needing repair; prolapse signals outlet obstruction and predicts secondary surgery.**
- The trap: confusing eversion with a true bladder diverticulum, and confusing prolapse of an intravesical ureterocele with a cecoureterocele, in which the ureterocele is **part of the urethra** rather than moving into it.

Endoscopic Evaluation

- Endoscopy is **the final element of evaluation — not always essential, but most cases come to operation and concurrent endoscopic evaluation is important.**
- **When endoscopic incision is to be performed, thorough endoscopy beforehand is critical to select the most effective site.**
- **Document the character of the urethra, bladder neck and trigone relative to the lesion, and the location of the other ureteral orifices.**
- **Seek the orifice of the affected ureter; if not seen in the bladder, examine the urethra carefully.**
- Technique as a sequence: **start with little bladder filling → increase volume slowly → watch the ureterocele flatten. Its true limits are best appreciated with limited filling, which exposes the lowest portion — probably the best site for incision.**
- **Detect extension into the urethra (cecoureterocele), which signals the need to ensure that section does not obstruct after incision.**
- **Retrograde contrast injection can confirm ureterocele disproportion — a narrow proximal ureter with no filling of an upper pole segment.**
- **Unusual connections with the genital ducts may be demonstrated — communication between ureter, vas deferens and seminal vesicles.**
- **An ectopic ureteral orifice may be very difficult to find endoscopically; if at the bladder neck it is often patulous.**
- **Some ectopic ureters have only a very thin layer of tissue separating them from the urethra, appearing more like a ureterocele; these may be associated with bladder neck defects and incontinence, and their**

endoscopic appearance supports a decision for **more extensive removal of the patulous segment with bladder neck reconstruction**.

Concept — *Endoscopy is not a diagnostic test here but a site-selection exercise: everything it adds — the lowest point of the ureterocele, urethral extension, a patulous bladder neck — exists to determine where to cut and what will be left obstructing afterwards.*

Concept — *Functional evaluation answers two separate questions with two separate tools: how much renal parenchyma is worth saving, answered by nuclear imaging only as "some" or "none", and how competent the bladder outlet is, answered by VCUG and videourodynamics — and it is the second, not the first, that predicts continence and the need for secondary surgery.*

KEY POINTS: EVALUATION OF ECTOPIC URETERS AND URETEROCELES

- **Ultrasound usually provides the anatomic diagnosis and permits an inference of renal function;** typical findings are hydroureteronephrosis in a single system, or a dilated upper pole with ureteric dilation in a duplicated system.
- The ureterocele is characterised by a **thin-walled cystic dilation within the bladder that does not extend beyond its walls**; a very dilated ectopic ureter impinging on the bladder and mimicking one is a **pseudoureterocele**.
- **Radionuclide imaging is the gold standard for renal functional assessment, best provided by DMSA**; a **MAG3 study with a loop diuretic** can be used where both function and degree of obstruction are needed.
- **The affected upper pole is the principal focus, but the health of the other moieties must be determined.**
- **VCUG provides the most definitive evaluation of the bladder and urethra, and the appearance of the bladder base with filling and voiding is useful in decision making — massive eversion indicates a weak trigonal floor more likely to require repair.**
- **Endoscopy should document the character of the urethra, bladder neck and trigone relative to the ureterocele or ectopic ureter.**

CLINICAL MANAGEMENT

Management Goals

- Although definitive management differs between the two entities, **the goals of management are identical and the approaches often are too.**
- **The goals of therapy: preservation of renal function; elimination of infection, obstruction and reflux; and maintenance of urinary continence.**
- **Minimisation of overall procedural morbidity is itself a goal, not an afterthought.**
- **The goals are agreed; the means of achieving them remain controversial.**
- One area of general agreement: **early institution of a daily prophylactic antibiotic reduces UTI risk in patients with significant hydronephrosis.**

Historical Perspective

COMPARISON — ERAS OF MANAGEMENT

Era	Development
Early 1950s	Open ureterocele excision described
Mid-1950s, revived 1970s	Total reconstruction — removal of non-functioning renal tissue plus bladder-level reconstruction with ureterocele excision and/or reimplantation
1980s	Upper tract approach became standard — upper pole partial nephrectomy to decompress the ureterocele
1990s	Endoscopic incision gained widespread favour , having begun as salvage in the septic child; now applied more selectively on imaging criteria
Recent	Observational management — controversial, best used for small ureteroceles with minimal hydronephrosis

- The table is for locating any given operation on a **single axis of decreasing invasiveness**, and for recognising that the older operations were not disproved so much as displaced.
- The trap: assuming each era abolished the last. **Total reconstruction, nephrectomy, incision and observation all remain in current use**, selected by patient rather than by date.
- For the **ectopic ureter**, historical management was **surgical in nearly all patients**, focused on **removal of the associated renal unit in patients with incontinence**, because **presentation was late and residual function usually limited** — so **excision was commonly the best option**.
- **With prenatal detection, salvage of the affected moiety has become a common option**, usually via an **upper-to-lower drainage procedure in duplicated systems**.

Concept — *The historical arc runs from removing the unit, to removing the pathology, to draining it, to watching it* — driven entirely by earlier detection, which changed the typical patient from a late-presenting child with a destroyed moiety to a neonate with potentially salvageable parenchyma.

Fetal Management

- Widespread fetal ultrasonography means a **large proportion are diagnosed prenatally**; duplicated systems **without hydronephrosis may not be noted**, but **hydronephrosis is usually easily seen**.
- **There is no role for fetal intervention in the ectopic ureter** — identification serves only to **plan postnatal management**.
- With prenatal suspicion: **obtain a postnatal ultrasound, consider a VCUG, and often start prophylactic antibiotics at birth** to limit early UTI risk.
- **Ureteroceles are likewise not an indication for fetal intervention, except when large and obstructing all renal units**.
- In those rare cases, **percutaneous needle aspiration of the ureterocele or cystoscopic laser ablation of the fetal ureterocele** has been reported — **still experimental and performed only in selected centres**.

Neonatal Management

Non-urgent Management

- **Most patients are seen non-urgently** — in antenatal consultation or as a neonate in clinic.
- Sequence: **evaluation always starts with renal ultrasonography** → if there is **significant hydroureteronephrosis** from either lesion, **consider starting prophylactic antibiotics and order a VCUG** → **decide on continuation of prophylaxis thereafter**.
- **Even in the absence of reflux, prophylaxis can be considered** to minimise infection risk with obstructed hydroureteronephrosis.

- A preoperative VCUG may be optional if the presence or absence of reflux would not change the decision to decompress endoscopically.
- A renal function study is often obtained once the child is old enough — generally older than 2 months — to assess function in the obstructed segment and other moieties; this too may not affect the initial decision.
- Small ureterocele, minimal hydronephrosis, no or little demonstrable obstruction → observation is a management option.
- Significant hydronephrosis with obstruction, especially with compromised parenchyma in the obstructed segment → proceed with TUI.
- Puncture methods: the most commonly used is a low transverse incision; the "watering can" technique uses a holmium laser to puncture multiple holes, in the hope of lower de novo VUR risk than traditional incision.
- If an endoscopic approach cannot be undertaken → cutaneous distal ureterostomy to divert urine until the child is old enough for formal reimplantation.
- The optimal age for elective endoscopic decompression is undefined, ranging from shortly after birth, to avoid infection, to several months, to reduce anaesthetic risk.

Urgent Management

- Child with sepsis of urinary origin and obstruction: antibiotics are usually sufficient to stabilise.
- If the child does not respond promptly and appropriately → TUI of an obstructing ureterocele, or nephrostomy tube drainage of the ectopic ureter.
- A cutaneous distal ureterostomy can be performed in either setting to establish drainage promptly and reliably.
- Once a child has proven prone to infection by presenting this way, a drainage procedure should be undertaken non-urgently after stabilisation to prevent future infections, even if not required acutely.

Concept — *The neonatal algorithm turns on one variable — is the segment obstructed and is the parenchyma at risk — with antibiotics buying time and a drainage procedure becoming obligatory once infection has declared the system vulnerable.*

General Management Principles

- For both lesions in a duplicated system, the principal concern is preservation of functional renal parenchyma, achieved by correcting obstruction and preventing reflux with its risk of infection-mediated parenchymal damage.
- The two goals can conflict: relieving obstruction may induce reflux in either or both poles; the same action may equally cause existing lower pole reflux to resolve.
- Decision making on when and how to intervene is quite subjective, and how much renal function is worth preserving must be decided individually.
- For the ectopic ureter: ureteral reimplantation, or in a duplex kidney common sheath reimplantation, or ureteroureterostomy, either low or proximal near the renal pelvis.
- Newer option for the young child not yet old enough for formal reimplantation: a side-to-side non-dismembered refluxing anastomosis with the bladder, performed in early infancy and intended as definitive in a population at low risk of UTI — long-term data are still required to establish safety and how many children need further procedures.
- For the ureterocele: TUI, or ureterocele excision with common sheath reimplantation, or ureteroureterostomy.

- **Acute decompression, by entity: ectopic ureter → end ureterostomy near the bladder; ureterocele → TUI is usually effective.**
- **Few objective criteria indicate how much residual function is worth preserving; VUR in either ureter must be considered, and every effort made to correct it with the drainage procedure.**
- **These lesions are overwhelmingly dealt with postnatally; fetal decompression is reported only for severe BOO from a ureterocele.**
- **Although earlier relief of obstruction might be expected to improve recovery of upper pole function, the evidence shows limited recovery even with early decompression. The counterpoint is an early endoscopic puncture series, which demonstrated function that had not been detected before drainage.**
- **Impaired upper pole function occurs equally in those diagnosed prenatally and postnatally.**
- **In most instances the upper pole contributes little to overall renal function, and there are no criteria for normal upper pole function.**

Concept — *Every operative choice in this chapter is a trade between obstruction and reflux across a moiety of unknown and unpredictable worth; because function cannot be predicted before or after decompression, the decision is empirical and individual, not algorithmic.*

KEY POINTS: CLINICAL MANAGEMENT — GOALS OF TREATMENT

- The goals are **preservation of renal function; elimination of infection, obstruction and reflux; and maintenance of urinary continence**, with **minimisation of overall procedural morbidity** as a further goal.
- For both entities in a duplicated system, a **primary concern is preservation of functional renal parenchyma.**
- **Decision making for parenchymal preservation is largely empirical**, with few objective criteria for how much residual function is worth preserving.
- For an ectopic ureter this can mean **common sheath reimplantation or ureteroureterostomy, either low or proximal near the renal pelvis.**
- For a ureterocele it can mean **TUI, or ureterocele excision with common sheath reimplantation or ureteroureterostomy.**
- **Observational management in carefully selected patients is reasonable**, with the potential for spontaneous decompression.

Observational Management

- Several groups report non-operative management of ureteroceles meeting criteria: **no obstruction of the ipsilateral lower pole or contralateral kidney and limited reflux to the lower pole (grade III or less); no function of the upper pole; or no obstruction on diuretic renography.**
- A described subset had **ureteroceles with multicystic dysplastic kidney in the associated upper segment, with low-grade or no reflux and no ureteric dilation; these prenatally diagnosed patients were managed non-operatively with a benign clinical course.**
- **Some children do well without intervention, and resolution of upper pole dilation and lower pole reflux has been reported.**
- Counselling is difficult: **the potential for later, unpredictable acute presentation is real**, and for that individual it is **clinically very significant even at a low statistical incidence.**
- **The long-term risk, the relative balance of morbidities and the robustness of clinical predictors are all undetermined.**

- **Carefully considered, observational management is an appropriate strategy with careful selection and parental education.**

Total Reconstruction

- Advocated by some as **the most definitive procedure** for ureteroceles.
- Components: **upper pole heminephrectomy + ureterocele excision + reimplantation of the lower pole ureter.**
- **Definitive, but an extensive operation performed through two incisions.**
- **Success rates are good, but it remains uncertain whether it is appropriate in most children when long-term relief from obstruction and reflux can be achieved with one, or at most two, lesser procedures.**
- **Best candidate: the older child with a massive ureterocele, a non-functioning upper pole, and significant lower pole reflux.**
- Contemporary execution: **partial nephrectomy laparoscopically or retroperitoneoscopically, with a lower Pfannenstiel incision for bladder reconstruction and reimplantation.**

Upper Pole Partial Nephrectomy

- **Long the preferred treatment where there is clearly no upper pole function, and where there is concern about how effective a drainage procedure will be because of massive dilation.**
- Numerous series advocate different approaches with results varying by **degree of reflux, age and underlying pathology**, but **no data indicate any one approach is definitively better as a universal choice.**
- **The surgical methods are similar whether open, conventional laparoscopic or robotic, and apply to both ectopic ureters and ureteroceles in duplex systems.**

Open Partial Nephrectomy or Heminephrectomy

- **A standardised procedure with relatively little recent evolution.**
- Access: **a flank approach usually offers better exposure to the upper pole vessels; dorsal lumbotomy is effective and less morbid than a muscle-dividing flank incision, but more challenging in older children.**
- **The primary concern is to avoid damaging the viable lower pole.**
- Sequence: **retract the kidney gently, avoiding vascular injury from traction and small-vessel spasm → transect the upper pole ureter and place a traction stitch on its proximal portion as a handle → dissect the ureter both below and above the vessels → pass the ureter behind the main renal vessels → dissect carefully around the renal vessels to protect the lower pole → sequentially ligate the upper pole vessels, most often two or three in number → demarcation of the upper pole parenchyma becomes apparent → resect the upper pole with electrocautery or another cutting device → strip the capsule off the upper pole in continuity for use in closure → close with several mattress sutures incorporating a pedicle of retroperitoneal fat.**
- Adjuncts: **atraumatic clamping of the renal pedicle allows a bloodless field; an intravenous osmotic diuretic such as mannitol immediately before and after clamping helps prevent acute tubular necrosis; topical vasodilators such as papaverine should be available for vasospasm. In general, vascular control is not essential, because the polar vessels can be identified and controlled.**
- **Upper pole ureterectomy: maintain the dissection immediately on the wall of the upper pole ureter to preserve the blood supply of the remaining lower pole ureter.**
- **If obstruction is accompanied by reflux into the ectopic ureter, some recommend a second (Gibson) incision to resect the ureter in its entirety; care must be taken to avoid injury to the vas deferens in males.**

- **To avoid common sheath complications, the back wall of the upper pole ureter can be left attached to the lower pole ureter, removing the remainder — this prevents damage to the lower pole ureteric blood supply, which courses between the two ureters.** Resection proceeds to the bladder, where sutures close the upper pole ureteral hiatus.
- **Alternative: ligate the distal refluxing stump and leave it in situ — few subsequent problems are attributed to the small volume of reflux. Larger-diameter stumps carry a higher risk of late infection. The refluxing, obstructed ectopic ureter raises the most concern, so remove as much ureter as possible without injuring the bladder neck.**
- **A Penrose drain, through a separate counterincision or at the incision edge, drains the renal fossa and the ureteral dissection.**
- **Postoperative evaluation with Doppler sonography to demonstrate normal anatomy, absence of a urinoma, and normal blood flow to the lower pole; a nuclear renal scan is optional if lower pole viability is in doubt.**

Laparoscopic and Robotic Partial Nephrectomy

- **Transabdominal or retroperitoneal, with robotic assistance increasingly common.**
- **Advantages: less postoperative pain, earlier return of gastrointestinal function, earlier discharge, quicker parental return to work; plus enhanced visualisation and magnification, improved cosmesis, and avoidance of the second incision often needed for distal ureterectomy.**
- **Feasible in very small infants, with operative time falling as experience and skill increase.**
- **Newer energy devices allow resection of the upper pole in a bloodless field.**
- **Transperitoneal sequence: grasp the pathological ureter as a handle and dissect closely on its wall to spare the normal ureter's blood supply → pass the upper pole ureter behind the vessels and use it to facilitate upper pole dissection → ligate the polar vessels with clips or divide with electrocautery, producing clearer demarcation → develop the plane between the upper pole collecting system and lower pole parenchyma bluntly → identify and transect the upper-to-lower pole attachments → remove the polar element with electrocautery or another energy source → check for collecting system leakage with intravenous methylene blue → close the defect over a pedicle of local fatty tissue.**
- **Closure of the defect may limit postoperative fluid collections; the described alternative is fibrin glue and haemostatic agents on the cut surface covered with Gerota fascia.**
- **Robotic advantages over standard laparoscopy: augmented magnification, greater instrument dexterity around the renal pedicle and polar vessels, and three-dimensional vision. Patient and port positioning are the same as for pyeloplasty.**
- **Vascular control may be by absorbable or metal clips or suture ties — ties may be preferable, reducing the risk of avulsing a clip during later dissection.**
- **The defect is usually closed with a tongue of retroperitoneal fat and two or three mattress sutures of 3-0 polydioxanone suture (PDS) or Vicryl.**
- **The robotic approach has been described for both upper and lower pole heminephrectomy with successful outcomes.**
- **Laparoendoscopic single-site surgery nephrectomy has been described for a single-system ectopic ureter, using one 22 mm multitrocar port recessed in the umbilicus for the entire procedure; single-site robotic platforms offer robotic precision with still less morbidity.**

Ureteral Clipping and Vascular Ligation

- **Laparoscopic clipping of the involved ureter without extirpative surgery has been described for non-functioning or poorly functioning moieties associated with ectopic ureter or obstructive ureterocele.**

- Reported complications included **pyonephrosis necessitating nephrectomy** and **febrile UTI with persistent upper pole hydronephrosis**.
- **The technique remains tentative and caution is advised; long-term follow-up is unavailable**, and patients should be **monitored long term for infection and hypertension**.
- **Laparoscopic vascular ligation of the poorly functioning upper pole vessels has also been reported as effective**.

COMPARISON — OPEN VERSUS MINIMALLY INVASIVE UPPER POLE NEPHRECTOMY

	Open	Laparoscopic / robotic
Access	Flank, giving best upper pole vessel exposure, or dorsal lumbotomy , less morbid but harder in older children	Transabdominal or retroperitoneal
Distal ureterectomy	Often needs a second (Gibson) incision	Second incision avoided
Vascular control	Polar vessels ligated sequentially; pedicle clamping optional with mannitol and papaverine	Clips, energy division or suture ties — ties preferred, to avoid clip avulsion
Defect closure	Capsule stripped in continuity, mattress sutures over a fat pedicle	Fat pedicle with 3-0 PDS or Vicryl mattress sutures, or fibrin glue with Gerota coverage
Distinctive advantages	Exposure and tactile control	Less pain, earlier gastrointestinal recovery, earlier discharge, magnification, cosmesis
Distinctive risk	Morbidity of the incision or incisions	Urinoma, especially where the polar defect is not formally closed

- The table is for choosing an **access route**, not an operation: the intrarenal steps are identical across all three platforms.
- The pattern: every difference between the rows is a difference of **exposure and closure**, and the one outcome difference that follows — urinoma — tracks **whether the polar defect was closed**, not which platform was used.
- The trap: reading the minimally invasive column as lower-risk overall. **The shared, dominant risk is loss of lower pole function, and it is unchanged by platform**.

Outcomes

- Results of upper pole removal for both entities are in general very good.
- For the ectopic ureter, the residual stump is rarely problematic.
- For ureteroceles with pre-existing lower pole reflux, resolution may be expected in up to 20%.
- New reflux appears in 15–50% of patients who had none preoperatively.
- The overall secondary surgery rate after primary upper pole nephrectomy for ureterocele is 40–50%.
- The literature is hard to interpret because indications for secondary surgery vary — in some series any postoperative reflux was an indication for bladder-level surgery, in others reflux was monitored and treated only for clinical problems such as UTI. **Neither approach has been validated by long-term data**.

Complications

- The most significant complication is loss of lower pole function.
- It may be a long-term issue and may not be recognised immediately; clinical signs — fever, increasing pain, haematuria — may appear in the first week after surgery.

- **Postoperative upper pole urinoma is reported in up to 20% of laparoscopic and robotic cases but is rarely of clinical significance; mostly in series with no formal closure of the polar defect. Whether they arise from injury to the lower pole or to the remnant upper pole is unclear. Asymptomatic, non-expanding urinomas can be observed.**
- **Less common problems: inferior vena cava laceration, duodenal perforation, need for total nephrectomy, and peritoneal tears if performed retroperitoneally.**

Concept — *Every technical rule in upper pole nephrectomy — the ureteric handle, the dissection kept on the ureteral wall, the back wall left attached, the avoidance of intravesical ureteric separation — exists to protect one structure: the longitudinal blood supply running between the two ureters and to the lower pole.*

Lower pole loss, not upper pole removal, is the operation's real risk.

KEY POINTS: UPPER POLE NEPHRECTOMY

- **The primary concern is to avoid damaging the viable lower pole.**
- **Separation of the duplicated ureters during intravesical dissection is discouraged**, because it injures the common blood supply running longitudinally between the two ureters.
- **Postoperative evaluation is best performed with Doppler sonography** — anatomy, urinoma, and blood flow to the lower pole.
- **Definitive cure by partial nephrectomy for ectopic ureters is usually successful**, the residual stump rarely being problematic.
- **For ureteroceles with lower pole reflux, resolution may be expected in up to 20%, and new reflux in 15% to 50% of those with none preoperatively.**
- **Complications include loss of lower pole function and postoperative upper pole urinoma.**

Lower Tract Reconstruction

- **A definitive reconstruction at the bladder is suitable for both the ectopic ureter and the ureterocele.**
- **Advantage: it relieves obstruction and corrects reflux in one operation.**
- **Disadvantages: potential injury to the bladder neck and vagina, and the complexity of the procedure.**
- **If clinically significant reflux persists after other procedures, lower tract reconstruction may be necessary.**

Ureterocele Excision and Common-Sheath Reimplantation

- **Sequence: transverse incision of the ureterocele between two stay sutures → develop a plane proximally between the ureterocele wall and the bladder wall → dissect the ureterocele off the bladder to the point where it joins the lower pole ureter → dissect the two ureters as a unit → taper the upper pole ureter as needed → reimplant both ureters submucosally → dissect the distal ureterocele in the same plane to the level of the bladder neck and resect it → plicate the detrusor if its backing appears insufficient to prevent reflux → raise bladder mucosal flaps to cover the area of the removed ureterocele.**
- **Separation of the duplicated ureters during intravesical dissection is discouraged** — it injures the common blood supply running longitudinally between the two ureters.
- **Plication of the detrusor underlying the ureterocele may be needed to reinforce areas of muscle deficiency.**
- **The distal portion of the ureterocele may extend below the bladder neck — extreme care is needed here to avoid injury to the sphincter mechanisms.**
- **If the entire ureterocele cannot be excised, it can be carefully fulgurated and closed over in two layers.**

- **Cecoureteroceles present a unique challenge:** the distal aspect can create an **obstructive flap-valve with voiding, acting like an obstructing windsock within the urethra** and causing bladder outlet obstruction. Options: **resection by gentle retraction if it is not very large; closure of the opening with two layers of tissue; or fulguration of the lumen to cause collapse and closure.** Careful postoperative assessment is needed to identify progressive voiding dysfunction if the remnant obstructs.

Marsupialisation

- **The thin intravesical aspect is removed and the edges oversewn, with no attempt to dissect the back wall or reinforce the posterior bladder wall,** on the empirical observation that this is not always needed. Reported results have been satisfactory.
- **Advantages:** avoids dissecting out the whole ureterocele and **avoids risk of injury to the underlying vagina.**
- **Potential risks:** diverticulum formation and altered bladder dynamics postoperatively in the wrong patient population.
- **Best for patients without a posterior detrusor defect, and perhaps smaller ureteroceles.**
- **The authors prefer complete excision with formal trigonal reconstruction** — definitively correcting the anatomical and functional defect at the first open operation. **If marsupialisation is used, it must be used with care.**
- **Results are very good, although persisting reflux can be an issue in 5–10% of patients,** more commonly when ureteral tapering is needed. Although its clinical significance has been questioned, **persisting reflux in this context must be viewed with caution.**

Concept — *Bladder-level reconstruction is the only procedure that addresses obstruction and reflux simultaneously, and it pays for that with proximity to the bladder neck, sphincter and vagina — which is why every technical caution in the section concerns what lies beneath and beyond the ureterocele, not the ureterocele itself.*

KEY POINTS: URETEROCELE EXCISION AND REIMPLANTATION

- **Cecoureteroceles present a unique challenge:** the distal aspect can create an **obstructive flap-valve with voiding, acting like a windsock in the urethra.**
- **Marsupialisation** is an alternative, in which the thin intravesical aspect is removed and the edge sutured.
- **Results of excision and common sheath reimplantation are very good, although persisting reflux occurs in 5% to 10%.**
- **Total reconstruction** — upper pole nephrectomy with ureterocele excision and lower pole reimplantation — **is definitive but extensive, requires two incisions, and may not be necessary in all patients.**

Pyeloureterostomy and Ureteroureterostomy

- **Indicated when the upper pole of either an ectopic ureter or ureterocele is to be maintained, because of function or surgeon preference.**
- **Increasingly used even in non-functioning upper poles, because concerns about long-term hypertension have not been substantiated and the procedure can be completed through a small incision, often no larger than a herniorrhaphy incision.**

Open Procedure

- **Distal ureteroureterostomy: end-to-side anastomosis between the upper pole ureter and the lower pole ureter**, performed via an **inguinal open approach**.
- It is **critically important to identify the recipient lower pole ureter correctly** — **cystoscopically place a stent into the lower pole ureter at the start of the procedure**.
- **Proximal alternative: ureteroureterostomy or pyeloureterostomy**, draining the upper pole system into the lower pole system. **High anastomoses may be preferable to a distal anastomosis with a dilated upper pole**, because the latter may produce more urinary stasis.
- The clinical significance of the "yo-yo" phenomenon remains uncertain, but a proximal anastomosis is preferred when there is significant ureteral size mismatch.
- Technique: **limit dissection to an absolute minimum, especially medially, to prevent disruption of either ureter's blood supply** → make a generous longitudinal pyelotomy or ureterotomy in the lower recipient pole to overcome size disproportion → anastomose end-to-side → aspirate the distal upper pole ureter with a feeding tube to decompress it → resect the distal upper pole ureter as far inferiorly as possible, staying directly on its wall to avoid the adjacent lower pole ureter's vasculature.
- If there is **no reflux**: resect as distally as possible and the remnant lower portion may be left open.
- If reflux is present in an ectopic ureter: take it as close to the bladder neck as possible.
- If associated with a ureterocele: resection of the ureterocele may be avoided as long as it is well decompressed.

Laparoscopic Procedure

- Both ureteroureterostomy and pyeloureterostomy can be readily performed laparoscopically, aided by **robotic control**, and are carried out at the level of the iliac vessels for the ectopic ureter presenting with incontinence.
- A stent is placed in the recipient ureter only, not across the anastomosis.
- **Robotic-assisted laparoscopic ureteroureterostomy (RALUU) versus open ureteroureterostomy**: comparable operative times and complication rates, with slightly shorter hospitalisation for robotic cases; in robotic cases **cystoscopy with lower pole stenting was consistently performed**, stent type varying by surgeon preference.
- A further RALUU series used an **external ureteral stent removed on postoperative day 1**, showing **complete resolution of hydronephrosis postoperatively**.
- **Laparoscopy-assisted open ureteroureterostomy**: an umbilical camera port for visualisation, then a Babcock passed through the small incision to grasp the ureters from the retroperitoneum and deliver them. No increase in operative time; success and complication rates comparable to the traditional procedure.
- Size mismatch between the two ureters does not seem to impede achieving a satisfactory anastomosis.

COMPARISON — DISTAL VERSUS PROXIMAL ANASTOMOSIS

	Distal ureteroureterostomy	Proximal ureteroureterostomy or pyeloureterostomy
Site	Distal ureter, inguinal approach	Near the renal pelvis
Recipient identification	Cystoscopic lower pole stent strongly recommended	Direct, at the pelvis
Preferred when	Ureters of similar calibre	Dilated upper pole; significant size mismatch
Drawback	More urinary stasis with a dilated upper pole	—

	Distal ureteroureterostomy	Proximal ureteroureterostomy or pyeloureterostomy
Stent	Recipient ureter only, never across the anastomosis	Same

- The table is for settling **where to site the anastomosis**; the operation is otherwise identical at either level.
- The pattern: every row reduces to **the calibre mismatch between upper and lower pole ureters**.
- The trap: assuming the distal route is simpler and therefore safer — **its specific hazards are recipient misidentification and stasis**.

KEY POINTS: PYELOURETEROSTOMY AND URETEROURETEROSTOMY

- Ureteroureterostomy has emerged as a **frequently used management technique for the duplicated collecting system with an obstructed upper pole, in both functioning and non-functioning upper poles**.
- **Size mismatch between the two ureters does not seem to impede achieving a satisfactory anastomosis**.

Transurethral Incision for Ureterocele

- The simplest means to decompress an obstructed upper pole or single-system ureterocele.
- Reported relief of obstruction: **78–97%**.
- TUI versus multiple punctures (watering can technique): **no difference in decompression rates, although multiple punctures are reported to induce less reflux into the affected ureter**.
- In acute sepsis TUI is the most appropriate intervention, though percutaneous nephrostomy is a viable option.
- The trade-off: **TUI reliably relieves obstruction but risks inducing reflux, which may predispose to upper tract infection and the need for reconstruction. The balance between TUI's minimal morbidity and its uncertainty is the basis of the clinical controversy about its utility**.
- Preferred method: **a transverse incision through the full thickness of the ureterocele wall using cutting current, made as distally on the ureterocele and as close to the bladder floor as possible to lessen the chance of postoperative reflux**.
- Instruments: **a Bugbee electrode or an angled-tip wire; the angled-tip wire is favoured — finer tip, more precision, and the angle allows easier manipulation. In older children, the resectoscope with the Collins hot knife may be used. Laser incision has equivalent results; cold knife incision or puncture has also been used**.
- The ureterocele should be incised deeply, because ureteroceles may be thick-walled.
- Adequacy is confirmed either by the escape of a jet of urine from the ureterocele or by viewing the urothelium on its inside.
- For an ectopic ureterocele extending into the urethra: **adequate drainage by either a longitudinal incision extending from the intravesical portion down into the urethral portion, or two separate punctures — one intravesical, one urethral. Retrograde incision of the intraurethral component has also been described for these more challenging cases**.
- Aftercare: **in the uninfected well child, no bladder catheter is left and most are treated as outpatients. Follow-up ultrasonography at 4 to 6 weeks assesses decompression; a voiding cystogram in the same period determines lower pole reflux status and whether new reflux into the ureterocele has been created**.

- **Reported incidence of new reflux after TUI: 0–50%, and the risk may depend on the method of incision.**
- **Several reports indicate equal decompression efficacy with less reflux using single or multisite puncture, and some authors therefore favour puncture over incision.**
- **The type of ureterocele is a clear determinant of TUI outcome: intravesical ureteroceles have the highest likelihood of achieving all therapeutic goals with incision or puncture alone.**
- **Decompression with no reflux is achieved in 70–80% of patients with an intravesical ureterocele and no pre-existing reflux.**
- **Extravesical ureteroceles are more likely to have persisting or new reflux and may require secondary surgery for reflux in up to 70% in some series.**
- **Meta-analysis: lower pole reflux and ureterocele location have similar impact on outcomes — relative risk of reoperation 1.74 times greater with reflux, and 2.78 times higher if extravesical. These factors were not additive, suggesting they represent similar underlying causes.**
- **Compared with any open approach, the simplicity of incision coupled with the potential for definitive cure — even in a minority — is appealing, and TUI better prepares the patient for secondary major surgery by decompressing dilated ureters.**

Reflux Outcomes After Transurethral Incision

- **The natural history of lower pole reflux after TUI is incompletely defined.**
- **An observational approach has been taken in some series, with resolution reported at variable follow-up.**
- **Cure — improvement of hydronephrosis with absence of VUR — was several times more likely in single-system than duplex-system ureteroceles, while de novo reflux into the ureterocele moiety was roughly twice as common in duplex systems.**
- **The usefulness of observing these patients depends on the occurrence of infection and on parental preference.**
- **Persisting reflux into the upper or lower pole is typically an indication for ureterocele excision and reimplantation, but resolution has been reported.**
- **In general it has not been possible to predict with any accuracy how an individual will respond to TUI from clinical parameters.**

COMPARISON — TRANSURETHRAL INCISION VERSUS WATERING-CAN PUNCTURE

	Transurethral incision	Multiple punctures ("watering can")
Technique	Transverse full-thickness incision with cutting current, as distal and as close to the bladder floor as possible	Multiple holmium laser punctures
Decompression	No difference between the two	No difference between the two
Reflux	Higher	Reported to induce less reflux
Evidence status	Long-established; the authors' preferred method	Several reports of equal efficacy with less reflux

- **The table isolates the single endpoint on which the two techniques differ — de novo reflux; decompression is equivalent.**

- The trap: treating the choice as settled. **The preferred technique in the chapter remains the transverse incision, despite the reflux signal favouring puncture.**

Concept — *TUI is not a lesser version of the definitive operations but a different kind of intervention altogether: an outpatient procedure that buys decompression and a real chance of cure, and even when it fails it improves the conditions for the operation that follows. Its outcome is predicted almost entirely by whether the ureterocele is intravesical and whether reflux is already present.*

KEY POINTS: TRANSURETHRAL INCISION OF URETEROCELE AND DRAINAGE PROCEDURES

- The reported incidence of new reflux after TUI is 0–50% and may depend on the method used for incision.
- The type of ureterocele is a clear determinant of outcome, with intravesical ureteroceles the most likely to achieve all therapeutic goals with incision alone.
- TUI also better prepares the patient for secondary major surgery, if needed, by decompressing dilated ureters.
- Persisting reflux into the upper or lower pole is typically an indication for excision and reimplantation, although resolution has been reported.
- Individual response to TUI cannot be predicted with accuracy from clinical parameters.
- The ectopic ureter in a neonate with sepsis or massive dilation may be best managed with a temporary end ureterostomy.
- Ureteroureterostomy can be performed distally or proximally, creating an end-to-side anastomosis between the upper and lower pole ureters.

Ureterostomy for Ectopic Ureter

- The ectopic ureter in a neonate with sepsis or massive dilation may be best managed with a temporary end ureterostomy, though endoscopic ureteral incision with stenting, or a nephrostomy tube, are alternatives.
- Diversion permits acute decompression to manage sepsis and allows later assessment of any function in the affected unit before definitive management.
- End ureterostomy allows the ureter to decompress and positions it for reimplantation if the unit proves salvageable.
- No resection of redundant ureter is performed — it will shorten in time, and length will be needed for reimplantation.
- The stoma should be positioned at what would be the lateral end of a Pfannenstiel incision.
- Percutaneous ureterostomy has also been reported.
- Assessment of functional status is performed shortly before definitive surgery, usually at 4 months or after 6 months of age. DMSA scanning is probably the most efficient means, with differential urine collection an option.
- If no function is appreciable, partial or total nephrectomy is performed with removal of the entire ureter to the stoma; a drainage procedure through ureteroureterostomy remains an option.
- Endoscopic incision of the bladder and ectopic ureter to create direct drainage into the bladder has been described, analogous to TUI of a ureterocele; open ureterovesicostomy for drainage has also been reported. Whether either is a long-term solution remains unclear.

Summary of Clinical Decision Making

Ectopic Ureter

- Decision making is much more straightforward than for ureteroceles and rests on whether to maintain the upper pole of a duplex system.
- If the upper pole is maintained, the approach depends on lower pole reflux:
 - Lower pole reflux present → common sheath reimplantation, or lower pole reimplantation with distal upper-to-lower ureteroureterostomy.
 - No lower pole reflux → either proximal or distal ureteroureterostomy.
- If the degree of function is ambiguous → a temporary end ureterostomy permits assessment after the acute presentation, particularly with a massively dilated ureter.
- If reflux is present, the need for lower tract surgery to correct it may influence the choice to avoid upper tract surgery altogether.
- If removal is chosen — for degree of dilation, or preference to remove non-functioning dysplastic tissue → upper pole nephrectomy.
- For the single-system ectopic ureter, preservation versus removal rests on degree of function and surgeon preference; there are no objective outcome data favouring either.

Ureterocele in a Duplex System

- The availability of endoscopic incision creates more options.
- TUI is an operation of a totally different nature from upper pole nephrectomy or lower tract reconstruction — it is inappropriate to equate them comparatively.
- Because significant lower pole reflux and upper tract dilation may in some patients be treated with a brief outpatient procedure with documented long-term results, it is reasonable to offer TUI before moving to complex upper or lower tract reconstruction.
- Even if half these patients need subsequent surgery, it can be safely deferred until the child is older.
- Further appeal: TUI may make subsequent surgery less complex by decompressing a dilated upper pole ureter — reimplantation may be much more effective and may not require excisional tapering, reducing overall morbidity.
- The appeal of TUI is less in the older child with a massive upper pole, in whom removal may be preferable and definitive surgery can be performed at diagnosis.

TABLE 52.1 OPTIONS FOR URETEROCELE MANAGEMENT

Procedure	Ideal indications	Advantages	Limitations
Transurethral incision	Infant; large ureterocele with VUR	Outpatient procedure; effective decompression; occasionally definitive	May produce reflux into the ureterocele segment, necessitating bladder surgery
Upper pole nephrectomy	Older patient; large non-functioning upper pole; no VUR	May be definitive; removes the pathology; avoids bladder-level surgery	May not be definitive; significant surgery; risk to the lower pole; may still require bladder surgery
Ureteroureterostomy or ureteropyelostomy	Older patient; functioning upper pole; no VUR	Drains the obstructed segment with little risk of obstruction or UTI	Leaves the ureterocele in the bladder; may develop reflux

Procedure	Ideal indications	Advantages	Limitations
Ureterocele excision and common sheath reimplantation	Reflux; functioning upper pole without significant dilation	Eliminates obstruction and reflux; removes the ureterocele; no renal risk	Complex surgery; risk to vagina and bladder neck; may require ureteral tapering

- **It is a selection grid, not a ranking:** the columns exist to match **age, upper pole function and the presence of VUR** to a procedure.
- **The pattern: VUR is the pivot. No VUR permits an upper tract solution** — nephrectomy or ureteroureterostomy; **VUR pushes towards a bladder-level solution** — excision and common sheath reimplantation — **except in the infant, where TUI is chosen precisely because it defers the bladder operation.**
- **Second pattern: upper pole function decides between nephrectomy and ureteroureterostomy** once an upper tract approach is chosen; **age decides between TUI and anything definitive.**
- **The trap:** reading the four rows as escalating degrees of the same operation. They are not comparable — **TUI is an outpatient endoscopic procedure whose failure costs little and may simplify the next operation**, whereas the other three are definitive operations with renal or bladder-neck risk.
- **Second trap:** inferring that ureteroureterostomy requires a functioning upper pole because the table says so — **the text states explicitly that it is now used even in non-functioning upper poles.**

Concept — *Decision making reduces to three variables — age, upper pole function, and the presence of reflux — and to one asymmetry: for the ectopic ureter the question is whether to keep the moiety; for the ureterocele the question is whether to start with the endoscope.*

Voiding Dysfunction After Ureterocele Repair

- **Voiding dysfunction after lower urinary tract reconstruction for ureterocele has been reported in some series, but with a low incidence; in other series there was no evidence of incontinence or significant bladder dysfunction.**
- **One series reported infrequent voiding with the possibility of infection in roughly half of patients, and incontinence in only a small minority;** a similar observation was made with a **lower incidence after complex bladder surgery for duplication anomalies.**
- **The bladder dysfunction has been attributed to an intrinsic bladder abnormality rather than to the surgical procedures.**
- **In the child with recurrent infections or incontinence after repair, consider the type of surgery and the possibility of inadequate trigonal support.**
- **Inadequate trigonal support leads to a weak bladder base, ballooning of the trigone posterior to the bladder neck, and an obstructive process** — with **inadequate bladder emptying, infection, and upper tract dilation.**
- **This can also occur after TUI of a ureterocele with a urethral component, as with a cecoureterocele.**
- **Assessment: VCUG to assess bladder anatomy with voiding, and videourodynamics if needed. Lateral views during voiding are essential to identify the bladder pathology. Antegrade cystoscopy via a suprapubic puncture has been used to better assess the bladder neck. Recurrent VUR may occur, particularly with obstructed voiding.**

- Management depends on the cause: **bladder and trigonal reconstruction or bladder neck repair; intermittent catheterisation; or endoscopic injection of bulking agents if there is bladder neck incompetence.**

Concept — *Post-repair voiding dysfunction is a trigonal support problem, not a sphincteric one* — which is why the diagnosis is made on lateral voiding views and the treatment is reconstruction of the bladder base rather than continence surgery alone.

OTHER URETERAL ANOMALIES

Anomalies of Number

Bifid Ureters and Ureteral Duplication

- Ureteral duplication may be associated with ectopia or a ureterocele but is equally compatible with a normally functioning renal system — provided both ureters enter the bladder orthotopically, or the duplication is partial.
- Incidence: approximately 1 in 125 people (0.8%) on autopsy series, which are less selective than clinical series.
- Slightly commoner in females, approximately 1.6:1.
- Unilateral duplication is six times more frequent than bilateral, but it is clinically prudent to look carefully for contralateral duplication whenever unilateral duplication is documented — this matters most in non-hydronephrotic upper poles and ureteral ectopia causing incontinence, where the second, occult system is the cause of persistent wetting after treating the first.
- Right and left sides are affected similarly in unilateral duplication.
- Definitional distinction: a bifid renal pelvis includes only a single ureter, whereas confluence of the upper and lower ureters below the ureteropelvic junction (UPJ) constitutes partial duplication, or bifid ureter.
- An increased incidence of ureterorenal pathology is documented with duplication anomalies, and the clinical implication depends largely on the ureteral insertion.

COMPARISON — THE ORGANISING CLINICAL RULE FOR DUPLICATION

	Lower pole	Upper pole
Governing mechanism	Conditions that routinely affect the single-system kidney	Conditions resulting from abnormal ureteral formation
Typical pathology	UPJ obstruction; vesicoureteral reflux	Ectopia; ureterocele
Management principle	As for the equivalent single-system lesion	As set out earlier in the chapter

- The table is the **single most useful clinical approach to duplication**: it predicts which disease will be found in which moiety before any imaging is reviewed.
- The trap: reversing it, and treating lower pole hydronephrosis as an ectopia-related problem.

Lower pole ureteropelvic junction obstruction

- Evaluation and management mirror single-system UPJ obstruction, and it occurs with both partial and complete duplication.

- **Recognition of duplication may be difficult with massive hydronephrosis, but the normal upper pole will usually be identified on functional imaging such as a renal scan, even when not seen on ultrasonography.**
- **It may resolve spontaneously, just as a single system with comparable dilation does.**
- **Optional repair: pyeloureterostomy of the dilated lower pole renal pelvis to the normal upper pole ureter.**
- **Upper pole UPJ obstruction has been reported, including alongside lower pole UPJ obstruction, but is exceptionally rare.**

Lower pole vesicoureteral reflux

- **Lower pole hydronephrosis may instead represent VUR, and this is often high grade.**
- **A dilated ureter is usually present on ultrasound; this rarely represents ureterovesical obstruction and more often indicates VUR.**
- **Lower pole VUR is often associated with significant functional reduction of the lower pole — usually congenital, but sometimes related to prior infection.**
- **Evaluation and management are as for reflux into a single system, but spontaneous resolution may be more delayed.**
- **Surgical options: common sheath ureteral reimplantation; endoscopic injection; or low ureteroureterostomy of the refluxing lower pole ureter to the upper pole ureter.**

Concept — *Duplication itself is not a disease: the moiety determines the pathology. The lower pole behaves like an ordinary kidney and acquires ordinary kidney problems; the upper pole, whose ureter budded abnormally, acquires the problems of abnormal ureteral formation. Note the mirror-image drainage operations — upper-to-lower for the obstructed upper pole, lower-to-upper for the refluxing lower pole.*

Triplication

- **Triplication of the ureter, complete or partial, is very rare, and typically presents with infection or wetting.**
- **The Smith classification divides triplex ureters into four types and remains useful.**
- **Orifice positioning typically follows the Weigert-Meyer law.**
- **Associations: ureteroceles; ectopia to the bladder neck, urethra or vagina; lower and mid-pole UPJ obstruction; ureterovesical obstruction; VUR; contralateral duplication; and fusion anomalies in some patients.**
- **Surgical management is tailored to the specific anatomy, and minimally invasive techniques have been reported effective.**

COMPARISON — SMITH CLASSIFICATION OF TRIPLEX URETERS

Type	Anatomy	Proportion
Type 1	Three entirely separate ureters, each with its own attachment to the bladder or distally	35%
Type 2	Incomplete separation with two ureteral orifices	21%
Type 3	Trifid ureter with a single ureteral orifice	31%
Type 4	Two ureters with three orifices — an inverted-Y bifurcation, as described for duplicated ureters	13%

- **The classification counts ureters against orifices, not ureters alone — that is the whole logic of the four types, running from three-and-three, through incomplete fusion, to three-and-one, and finally the anomalous three-orifices-from-two-ureters.**

- The trap is assuming the numbering runs in order of frequency: it does not — type 1 is commonest, but type 3 outranks type 2.

Quadruple Ureters

- Ureteral quadruplication is rarer still, and most reported cases have been in adults.
- Three recent cases involved four ureters draining into a large ureteral cyst and connecting to the bladder through a single ureter.
- One report of a young adult with incontinence described three of four ureters merging and entering the bladder orthotopically, with the fourth draining to the perineum — and the ectopic ureter arose from the lower mid-calyx, contrary to the prediction of the Weigert-Meyer law.

Concept — As ureteral number increases, the Weigert-Meyer law degrades from rule to tendency — it still typically holds in triplication and is explicitly violated in quadruplication. The law describes the ordering of two sequential buds; it has no authority over anatomy generated by additional, unordered budding events.

Fibroepithelial Polyps

- **Presentation:** flank pain, haematuria, or incidental detection of hydronephrosis.
- The commonest site of attachment is the UPJ, although they may originate from any part of the ureter.
- As a cause of UPJ obstruction they are very uncommon — reported in 0.5% of all UPJ patients undergoing pyeloplasty.
- A very large majority of patients are male (89%), and a majority of polyps are left-sided (78%).
- Bilaterality is uncommon but reported, with clinical effects on both kidneys; polyps may also be multiple, and protrusion from the urethra and from the ureteral orifice has been published.
- Smaller polyps may not be evident on modern imaging: only 22% were detected before surgery for UPJ obstruction.
- **Ultrasonography may be useful in detection.** They would usually be seen on modern renal imaging but may be mistaken for ureteral blood clots, particularly in the setting of haematuria.
- A filling defect on retrograde pyelography performed in anticipation of pyeloplasty should trigger a search for polyps — though this is unlikely to alter the surgical approach unless the polyp lies significantly distal to the UPJ.
- **Aetiology is unclear;** progressive traction from ureteral peristalsis may promote oedema and growth. One case is described in which the ureter was intussuscepting into itself from traction on a mid-ureteric polyp. An indwelling nephrostomy tube has also been reported as a cause.
- **Histology:** benign neoplasms with fibroepithelial and vascular elements, overlying normal to hypertrophied urothelium, usually with a fibrovascular core and significant stromal oedema. Little inflammation is seen.

COMPARISON — FIBROEPITHELIAL POLYP MANAGEMENT BY SITE

	Polyp at the UPJ	Polyp away from the UPJ
Recommended treatment	Pyeloplasty in most reported cases	Ureteroscopic removal — the recommended therapy
Status of the alternative	Whether simple endoscopic resection would suffice is unsettled — too few cases	Established on combined adult and paediatric reports

	Polyp at the UPJ	Polyp away from the UPJ
Rejected historical options	Sleeve resection and reanastomosis to prevent recurrence — persisting success with ureteroscopic resection suggests it is unnecessary; nephrectomy, recommended in still earlier reports, cannot be justified	

- The table turns on **whether the UPJ itself must be reconstructed**, which is the only question the polyp's location actually settles.
- The trap: escalating treatment for fear of recurrence, when **the historical escalation to sleeve resection and nephrectomy has not been justified by outcomes**.

Concept — *The fibroepithelial polyp is a benign lesion whose entire management question is whether the UPJ itself must be reconstructed — at the UPJ the obstruction is treated, not merely the polyp; elsewhere the polyp alone is removed endoscopically.*

Anomalies of Position

Vascular Anomalies Involving the Ureter

- A variety of vascular lesions can cause ureteral obstruction, and in all of them **the vascular system, rather than the urinary system, is the anomalous one**.
- With the exception of accessory renal blood vessels, all are relatively uncommon, although all have clinical relevance.

Preureteral Vena Cava (Circumcaval or Retrocaval Ureter)

Terminology

- Commonly known to urologists as circumcaval or retrocaval ureter — terms that are **anatomically descriptive but misleading as to development**.
- "Preureteral vena cava" is the more accurate term, because it **emphasises that the lesion results from altered vascular, not ureteral, development**.

Anatomy

- Involves the right ureter, which deviates medially behind (dorsal to) the inferior vena cava, winds about it and crosses in front of it from medial to lateral, then resumes a normal distal course to the bladder.
- The renal pelvis and upper ureter are typically elongated and dilated in a J or fishhook shape before passing behind the cava.
- The collecting system is not inevitably obstructed.

COMPARISON — THE TWO CLINICAL TYPES OF CIRCUMCAVAL URETER

	Type I — the commoner	Type II
Hydronephrosis	Present, with a typically obstructed pattern	Lesser degree, or none at all
Ureteric configuration	Fishhook-shaped deformity to the level of the obstruction	Upper ureter not kinked; pelvis and upper ureter lie almost horizontal, encircling the cava in a smooth curve
Level of crossing	Lower	Passes behind the cava at a higher level

	Type I — the commoner	Type II
Site of obstruction	At the edge of the iliopsoas muscle, where the ureter deviates cephalad before passing behind the cava	Not obstructing

- The table separates **an anatomical variant from an obstructive lesion**: both have the same venous embryology, and only type I needs an operation.
- The trap: assuming that a circumcaval course implies obstruction. **The level and angle of crossing, not the crossing itself, determine whether the ureter obstructs.**

Embryology

- The **definitive inferior vena cava develops on the right from a plexus of fetal veins**. Retroperitoneal venous pathways are initially symmetrical: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally, forming a collar on each side through which the ascending kidneys pass.
- **Normal course**: the left supracardinal veins and the lumbar portion of the right posterior cardinal vein atrophy → the subcardinal veins become the internal spermatic veins → the definitive right-sided inferior vena cava forms from the right supracardinal vein.
- **The anomaly**: if the lumbar portion of the subcardinal vein fails to atrophy and becomes the primary right-sided vein, the ureter is trapped dorsal to it.
- **Double right vena cava**: when the definitive cava forms normally and the ventral portion of the primitive ring also persists, both the dorsal and the ventral right-sided components survive, and the right ureter is trapped between the two limbs.
- **Bilateral or left-sided vena cava** can occur, and a bilateral circumcaval ureter has been described in situs inversus. In bilateral vena cava with a circumcaval ureter, the circumcaval ureter has been reported only on the right — indicating the right cava developed abnormally from a persistent subcardinal vein while the left developed normally from the left supracardinal vein.

Incidence and presentation

- Autopsy incidence about 1 in 1500 (0.07%); a recent meta-analysis indicated a prevalence of 0.13%.
- About three times commoner in male than female cadavers.
- The symptoms are those of obstruction. Although the lesion is congenital, most patients do not present until the third or fourth decade.

Diagnosis

- Flank or abdominal pain, or infection — or incidental discovery during other radiological tests.
- Excretory urography often fails to visualise the ureter beyond the J hook, the segment extending behind the cava.
- Retrograde ureteropyelography demonstrates an S curve to the point of obstruction, with the retrocaval segment lying at the level of L3 or L4.
- Ultrasonography, CT and MRI have all been useful in defining the vascular malformation.
- CT urography may be the procedure of choice, confirming the diagnosis and avoiding retrograde ureteropyelography.
- Nuclear renal furosemide scanning can assess the severity of obstruction but may not identify the anomaly itself.
- MRI can demonstrate the course of the preureteral cava and may be more detailed and less invasive than CT and retrograde pyelography.

- **The diagnosis may not be obvious:** a case is reported in which obstruction from a preureteral cava was taken for UPJ obstruction, and the open pyeloplasty neither recognised the true cause nor succeeded. Reoperative robotic correction of a missed preureteral cava after pyeloplasty has been described, underlining that modern imaging does not always permit an accurate diagnosis.

Treatment

- **Surgical correction:** ureteral division, with relocation and ureteroureteral or ureteropelvic reanastomosis, usually with excision or bypass of the retrocaval segment, which can be aperistaltic.
- **Be mindful of the ureteric blood supply** — from the renal artery and aorta superiorly and the iliac vessels inferiorly.
- **The preferred approach for the obstructed ureter is ureteral division and relocation.**
- **Laparoscopic and robotic reconstruction, by both transperitoneal and retroperitoneal approaches, has been described in children.**

Concept — *The name carries the teaching: the cava is the anomaly, the ureter is merely trapped. That explains why the lesion is right-sided, why it presents in adulthood despite being congenital, why the retrocaval segment is excised or bypassed rather than simply released, and why it is the classic mimic of UPJ obstruction that a pyeloplasty will fail to cure.*

Other Anomalies of Position

- **Several instances of horseshoe kidney have been reported** in association, as have a **variety of left renal anomalies** — agenesis, hydronephrosis, malrotation and hypoplasia.
- **Mimics of circumcaval ureteral obstruction:** an obstructing branch of the right spermatic vein, a lumbar vein, and an anomalous tendon of the iliopsoas muscle.

Preureteral iliac artery (retroiliac ureter)

- **A ureter coursing behind the common iliac artery is rare;** either side may be involved, and **bilateral cases have been reported.**
- **Obstruction occurs at the level of L5 or S1 as the ureter is compressed behind the artery.**
- **Coexisting anomalies are common, particularly vasal anomalies.**
- **Like the preureteral cava, it is considered to be of vascular origin without definitive proof.**
- **Embryology:** normally the **primitive ventral root of the umbilical artery is replaced by a more dorsal branch developing between the aorta and the distal umbilical artery;** persistence of the ventral root while the dorsal root fails to form traps the ureter dorsally.
- **Ureteral or mesonephric duct ectopia is often present.** Reported combinations include **an ectopic ureter emptying with the ipsilateral vas deferens via a persistent common mesonephric duct into the proximal posterior urethra;** **bilateral retroiliac ureters with bilateral ectopic termination of the vasa deferentia into the ureters;** and **ectopic vaginal termination with urometrocolpos from an imperforate hymen.**
- **A retro-internal iliac artery ureter causing left ureteral obstruction** has been reported in an otherwise normal young female.

Vascular obstruction of the distal ureter

- **Obstruction of the distal ureter by uterine, umbilical, obturator and hypogastric vessels close to the bladder has been described.**
- **It is not always clear that vascular impressions on a dilated ureter are causal** — the finding may be an artefact when a ureter dilated by intrinsic obstruction is secondarily compressed against an adjacent vessel.

- Judging from the paucity of contemporary reports, primary terminal ureteral obstruction by vascular lesions is likely to be rare.

Herniation of the ureter

- **Extremely rare.** The classic series covered **inguinal, scrotal and femoral herniations**.
- **Most were paraperitoneal** — a loop of herniated ureter alongside a peritoneal hernial sac; only a minority were extraperitoneal, with no hernial sac.
- In paraperitoneal ureteral hernias, the ureteric loop is always medial to the peritoneal sac.
- Of six scrotal hernias, four had no peritoneal sac. When the ureter extended into the scrotum it was more likely to be dilated, causing upper tract obstruction.
- In children, herniated ureters have presented with hydronephrosis, associated with megaureters and with persistent hydronephrosis after posterior urethral valve ablation.
- Internal hernias of the ureter are even more exceptional: reports exist of a sciatic hernia containing a ureter, herniation between the psoas muscle and the iliac vessels, and lumbar triangle herniation.
- Ureteral herniation with obstruction has been reported as a rare complication of renal transplantation.

COMPARISON — THE TWO NAMED PREURETERAL VASCULAR LESIONS

	Preureteral vena cava	Preureteral iliac artery
Side	Right, even when bilateral cavae exist	Either; bilateral reported
Level of obstruction	L3–L4 for the retrocaval segment; type I obstructs at the iliopsoas edge	L5–S1
Embryological error	Lumbar subcardinal vein fails to atrophy and becomes the definitive cava	Ventral root of the umbilical artery persists while the dorsal root fails to form
Characteristic imaging	J or fishhook pelvis; S curve on retrograde study; non-visualisation beyond the J hook on excretory urography	Compression at the arterial crossing
Associated anomalies	Horseshoe kidney; left renal anomalies	Vasal and mesonephric duct anomalies; ureteral ectopia
Treatment	Division, relocation, reanastomosis with excision or bypass of the aperistaltic segment	Tailored, on the same principle of relocating the ureter

- The table's organising row is **level of obstruction**, which alone distinguishes the two lesions radiologically.
- The pattern: both are **failures of regression in a paired embryonic vessel**, and both are treated by moving the ureter rather than the vessel.
- The trap: expecting the arterial lesion to be right-sided by analogy with the venous one — **only the cava lesion is obligately right-sided**.

Concept — Every lesion in this section shares one mechanism — **a vessel that should have regressed did not, and the ureter was caught on the wrong side of it.** The treatment principle follows directly: move the ureter, not the vessel, and excise the segment rendered aperistaltic by the entrapment.

KEY POINTS: OTHER URETERAL ANOMALIES

- Ureteral duplication is **compatible with a normally functioning renal system if both ureters enter orthotopically or the duplication is partial**.
- It is clinically prudent to look for contralateral duplication when unilateral duplication is documented.

- **An increased incidence of ureterorenal pathology is documented with duplication anomalies.**
- **Conditions routinely affecting a single-system kidney affect the lower pole — UPJ obstruction and VUR; the upper pole is more likely affected by conditions of abnormal ureteral formation — ectopia and ureterocele.**
- **Polyps of the ureter may present with flank pain, haematuria or incidental hydronephrosis, and the commonest site of attachment is the UPJ.**
- **UPJ polyps have mostly been treated with pyeloplasty, although whether endoscopic resection would suffice is unsettled; ureteroscopic removal is the recommended therapy away from the UPJ.**

KEY POINTS: PREURETERAL VENA CAVA

- **The lesion involves the right ureter, which deviates medially behind the inferior vena cava, winds about it and crosses in front from medial to lateral before resuming a normal course to the bladder.**
- **If the lumbar subcardinal vein fails to atrophy and becomes the primary right-sided vein, the ureter is trapped dorsal to it.**
- **It may present with flank or abdominal pain or infection, or be discovered incidentally.**
- **Surgical correction involves ureteral division, with relocation and ureteroureteral or ureteropelvic reanastomosis, usually with excision or bypass of the aperistaltic retrocaval segment.**